Topic: The Laplace transform

Question: Find the Laplace transform, given $s > -1$.

$$\mathscr{L}(e^{-t})$$

Answer choices:

A $F(s) = \dfrac{1}{-s+1}$

B $F(s) = \dfrac{1}{-s-1}$

C $F(s) = \dfrac{1}{s+1}$

D $F(s) = \dfrac{1}{s-1}$

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Solution: C

We're transforming $f(t) = e^{-t}$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty e^{-st} e^{-t}\ dt$$

$$F(s) = \int_0^\infty e^{-st-t}\ dt$$

$$F(s) = \int_0^\infty e^{(-s-1)t}\ dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = \frac{1}{-s-1} e^{(-s-1)t} \Big|_0^\infty$$

$$F(s) = \lim_{t\to\infty} \left(\frac{1}{-s-1} e^{(-s-1)t} \right) - \frac{1}{-s-1} e^{(-s-1)(0)}$$

$$F(s) = \lim_{t\to\infty} \left(\frac{1}{-s-1} e^{(-s-1)t} \right) + \frac{1}{s+1}$$

If we evaluate the limit assuming $-s-1 < 0$, or $s > -1$, the exponential tends toward 0 and the Laplace transform is

$$F(s) = \frac{1}{s+1}$$

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Topic: The Laplace transform

Question: Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(3)$$

Answer choices:

A $F(s) = -\frac{3}{s}$

B $F(s) = \frac{3}{s}$

C $F(s) = -\frac{1}{3s}$

D $F(s) = \frac{1}{3s}$

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Solution: B

We're transforming $f(t) = 3$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty 3e^{-st}\ dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = \frac{3}{-s} e^{-st} \Big|_0^\infty$$

$$F(s) = \lim_{t \to \infty} \left(\frac{3}{-s} e^{-st} \right) - \frac{3}{-s} e^{-s(0)}$$

$$F(s) = \lim_{t \to \infty} \left(\frac{3}{-s} e^{-st} \right) + \frac{3}{s}$$

If we evaluate the limit assuming $s > 0$, the exponential tends toward 0 and the Laplace transform is

$$F(s) = \frac{3}{s}$$

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Topic: The Laplace transform

Question: Find the Laplace transform, given $s > 0$.

$$\mathscr{L}(t)$$

Answer choices:

A $F(s) = \dfrac{1}{s^2}$

B $F(s) = \dfrac{1}{s}$

C $F(s) = \dfrac{1}{s} + \dfrac{1}{s^2}$

D $F(s) = \dfrac{1}{s+1}$

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Solution: A

We're transforming $f(t) = t$, so we'll plug this into the definition of the Laplace transform.

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty e^{-st} t\ dt$$

$$F(s) = \int_0^\infty te^{-st}\ dt$$

Now we can integrate and evaluate over the interval.

$$F(s) = -\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\Bigg|_0^\infty$$

$$F(s) = \lim_{t\to\infty}\left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right) - \left(-\frac{(0)e^{-s\cdot 0}}{s} - \frac{e^{-s(0)}}{s^2}\right)$$

$$F(s) = \lim_{t\to\infty}\left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right) + \frac{1}{s^2}$$

If we evaluate the limit assuming $s > 0$, the exponential tends toward 0 and the Laplace transform is

$$F(s) = \frac{1}{s^2}$$

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